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GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

214138.1.pdf *Geochronology Record 1905*

214138.1: granodiorite gneiss, Joker Well

(Goonetarra Granodiorite, Rothsay Suite, Annean Supersuite; Youanmi Terrane, Yilgarn Craton)

IOH Fielding, Y Lu, MTD Wingate and I Zibra

Location and sampling

YALGOO SH50-02, YALGOO 2241 Zone 50, MGA 474384E 6850813N
Warox Site AETYLG000128 Sampled on 28 May 2014

This sample was collected from a small pavement on Bunnawarra Station, about 22.3 km northeast of Bunnawarra Homestead, 15.2 km northwest of Badja Homestead, and 5.0 km southwest of Joker Well.

Geological context

The unit sampled is the Goonetarra Granodiorite of the Rothsay Suite, Annean Supersuite (Van Kranendonk et al., 2013; Zibra, 2020: ENS Report Goonetarra Granodiorite, viewed 28 April 2023). The 2763–2749 Ma Goonetarra Granodiorite consists mainly of metatonalite, metagranodiorite, and metamonzogranite, which are typically foliated and lineated (Zibra, 2020: ENS Report Goonetarra Granodiorite, viewed 28 April 2023). In the Yalgoo dome the Goonetarra Granodiorite is intruded into the Kynea Tonalite and into the greenstone succession (Zibra et al., 2018; Clos et al., 2018). Near the centre of the Yalgoo dome, the Goonetarra Granodiorite is intruded by monzogranite of the 2650–2600 Ma Walganna Suite of the Bald Rock Supersuite (Zibra, 2020: ENS Report Goonetarra Granodiorite, viewed 28 April 2023). Two monzogranite gneisses of the Goonterra Granodiorite sampled about 90 m to the northeast and 1.5 km to the east-southeast yielded igneous crystallization ages of 2761 ± 4 Ma and 2763 ± 6 Ma, respectively (GSWA 214139, Fielding et al., 2023b; GSWA 214101, Lu et al., 2016).

Petrographic description

The sample is a fine-grained, equigranular granodiorite gneiss, consisting of about 40–45% plagioclase, 25% quartz, 20–25% K-feldspar, 5% biotite, 2–3% muscovite, 1–2% opaques and trace myrmekite, apatite, allanite and zircon. Plagioclase is partially sericitised and forms anhedral, equant to tabular grains up to 3 mm long. Plagioclase porphyroclasts are wrapped by elongate trails of biotite and quartz ribbons, defining a metamorphic fabric. Quartz is anhedral, weakly strained and commonly occurs as irregular aggregates intergrown with plagioclase and K-feldspar. K-feldspar is anhedral, <1mm in size, and exhibits coarse microcline twinning and some contain wisps of flame perthite. Myrmekite lobes, up to 0.5 mm in diameter, are in contact with K-feldspar. Biotite forms as randomly oriented, subhedral to anhedral laths up to 1.5 mm long and is partly replaced by chlorite and epidote. Muscovite is anhedral, <1mm long, and occurs as either single grains or clusters replacing plagioclase. Probable magnetite grains are cubic and up to 0.5 mm in diameter. Apatite is euhedral and up to 0.2 mm long. Zircon within biotite are surrounded by dark haloes. Allanite is rare and mostly replaced by epidote. Microstructures indicate that fabric developed under mid- to low-amphibolite facies conditions.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, and subhedral to euhedral. The crystals are up to 400 µm long, and elongate, with aspect ratios up to 5:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. Many crystals contain high uranium, metamict zones. Some crystals appear to contain older cores. A CL image of representative zircons is shown in Figure 1.

Analytical details

This sample was analysed on 18–19 May 2015, using SHRIMP-B. Ten analyses of the BR266 standard were obtained during the session. Significant drift of standard $^{238}U/^{206}Pb^*$ dates during the session was addressed by fitting a LOWESS curve (Cleveland, 1979) with a smoothing window of six analyses, implemented using the program Isoplot 2.50 (Ludwig, 2009; Fielding et al., 2023a). During the session, ten standard analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.10% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eleven analyses were obtained from nine zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 2).

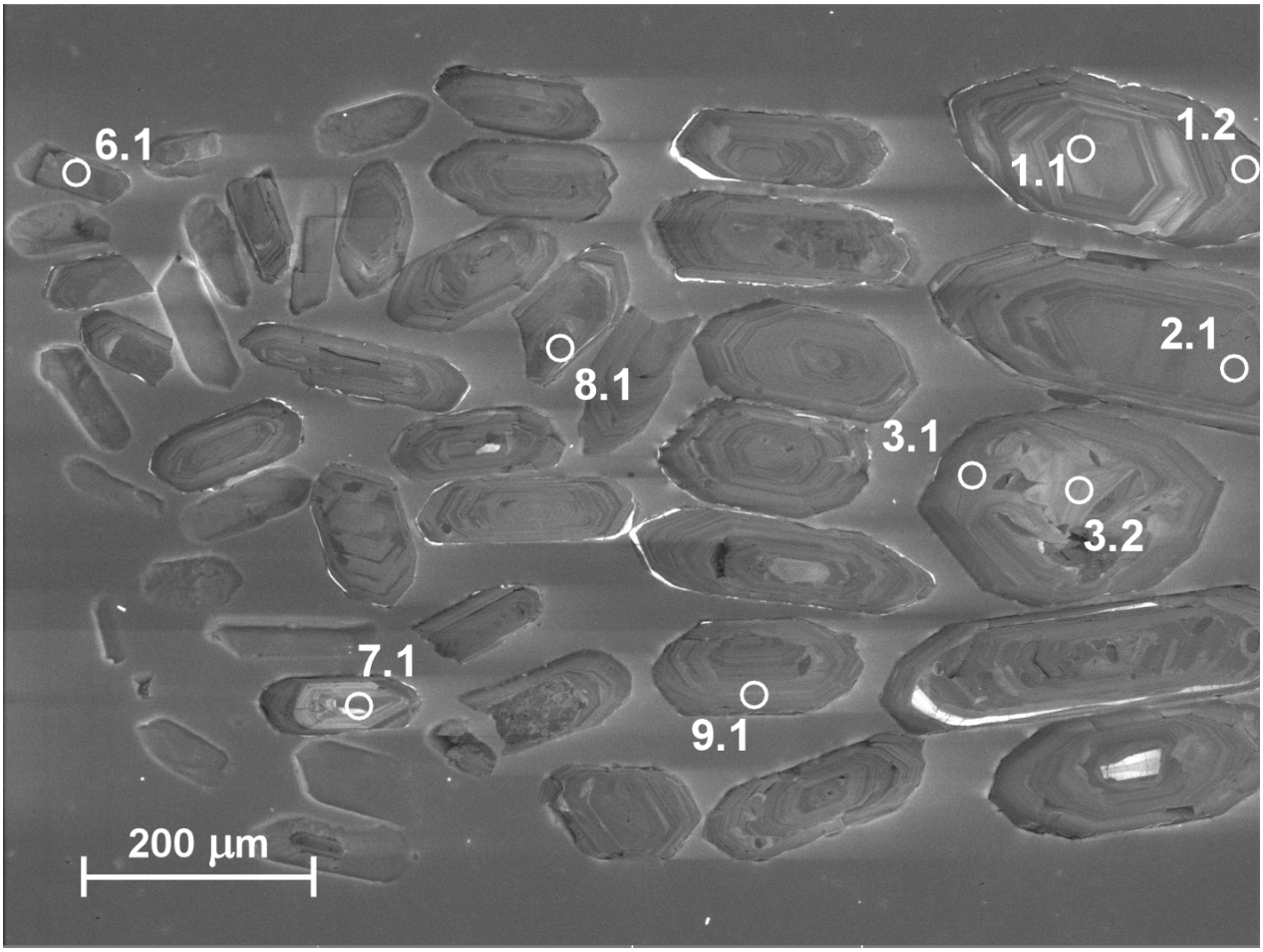

Figure 1. Cathodoluminescence image of representative zircons from sample 214138: granodiorite gneiss, Joker Well. Numbered circles indicate the approximate locations of analysis sites

Interpretation

The analyses are concordant to slightly discordant (Fig. 2), and form two groups, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

Group I comprises nine analyses of nine zircons (Table 1), which yield a weighted mean $^{207}Pb^{*}/^{206}Pb^{*}$ date of 2763 ± 5 Ma (MSWD = 0.34).

Group X comprises two analyses of two zircon cores (Table 1), which yield a weighted mean $^{207}Pb^{*}/^{206}Pb^{*}$ date of 2784 ± 13 Ma (MSWD = 0.02).

The date of 2763 ± 5 Ma for the nine analyses in Group I is interpreted as the magmatic crystallization age of the granodiorite protolith.

The date of 2784 ± 13 Ma for the two analyses in Group X is interpreted as the age of an inherited component.

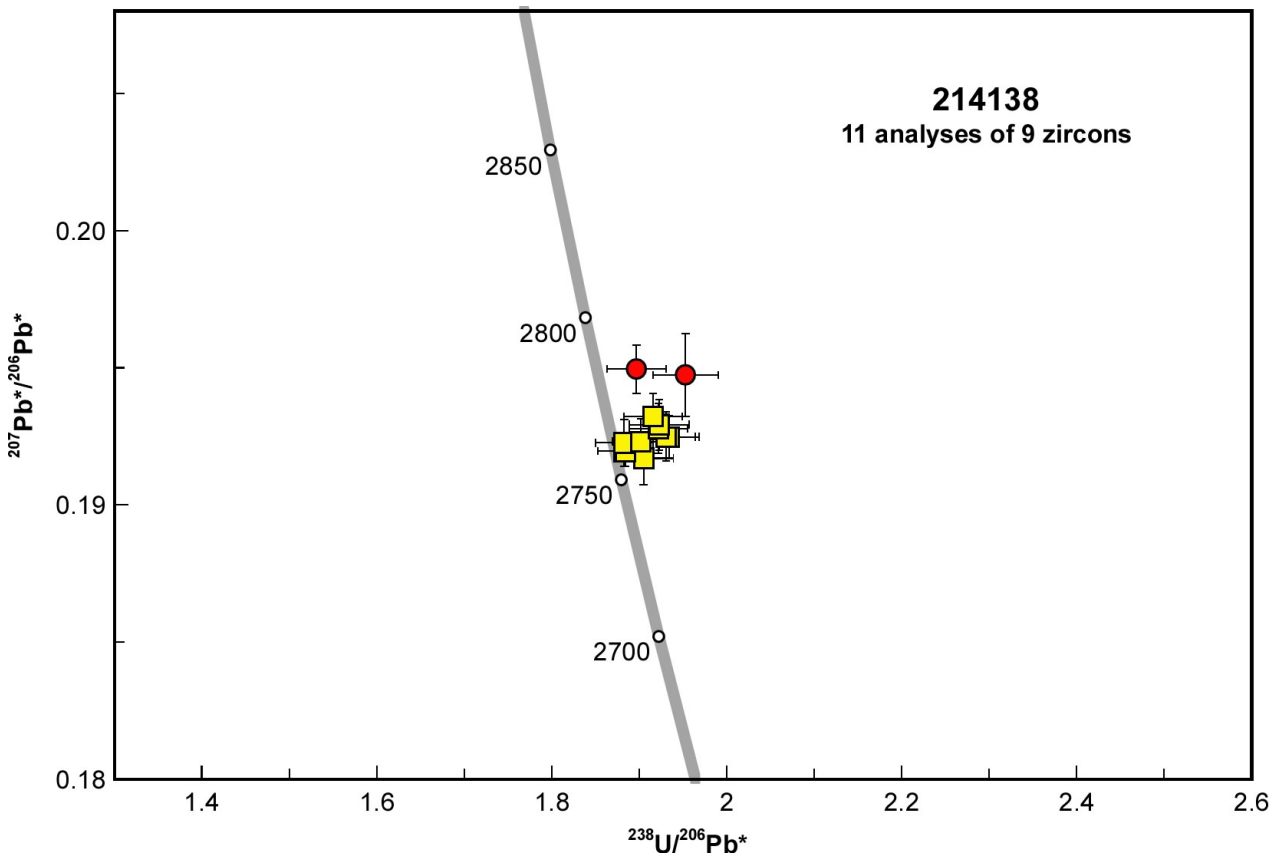

Figure 2. U–Pb analytical data for zircons from sample 214138: granodiorite gneiss, Joker Well. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircon cores)

Links

Table 1. Ion microprobe analytical results for zircons from sample 214138: granodiorite gneiss, Joker Well: 214138.1.Table1.xls
SHRIMP spot images: 214138_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

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Recommended reference for this publication

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

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Information Centre
Department of Mines, Industry Regulation and Safety
100 Plain Street
EAST PERTH, WA 6004
Telephone: +61 8 9222 3459 | Facsimile: +61 8 9222 3444
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